

AI agent harness: What makes agents reliable in 2026

Hill-Top Consultancy Expert Blog

Author: Leon Okafor, Principal AI Engineer

Published: 15 September 2026

Reading time: 11 min read

In perspective

An AI agent harness is the engineering system around an agent that makes its behavior observable, testable, and safe to operate. It includes tools, context, policies, output checks, and recovery paths.

The five control surfaces

A useful harness controls identity, tools, context, evaluation, and recovery. Identity determines what the agent is allowed to do. Tools determine how it can act. Context determines what it knows. Evaluation determines whether the result is acceptable. Recovery determines what happens when the plan fails.

Reliability is a system property

A capable model can still produce an unreliable agent if tool responses are ambiguous, state is lost, permissions are too broad, or failures are invisible. Reliability comes from typed tool contracts, explicit state transitions, idempotent actions, timeouts, and human review for consequential steps.

A practical evaluation loop

Build a scenario set from real work. Capture the input, plan, tool calls, intermediate decisions, final result, and human correction. Evaluate task completion, policy compliance, efficiency, and user trust.

Key takeaways

- Design the harness before increasing agent autonomy.

- Make every tool call inspectable and permissioned.
- Evaluate complete trajectories, not only final answers.
- Keep a human path for uncertainty and high-impact actions.

[Contact Hill-Top Consultancy](#)